

Career Intelligence Bot: An AI-Powered Platform for Job Search Optimization Using LLMs, RAG, and Agentic Design

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Final Project Report

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GitHub: https://github.com/YOUR_USERNAME/career-intelligence-bot
Video Demo: https://youtu.be/YOUR_VIDEO_ID

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Abstract

Job searching remains a time-intensive, opaque process where candidates lack visibility into how their qualifications match market demands. This paper presents the **Career Intelligence Bot**, an AI-powered platform that integrates Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), and agentic AI design to provide comprehensive career guidance. The system performs real-time resume analysis, job market intelligence gathering, ATS (Applicant Tracking System) simulation, and personalized career roadmap generation. Unlike traditional job boards, this platform provides dual-score analysis (ATS + Recruiter probability), weighted skill matching against actual job descriptions, seniority signal detection, and business impact quantification. The system demonstrates practical applications of eight distinct AI/ML topics including prompt engineering, responsible AI, and explainable AI. Evaluation shows the platform successfully identifies skill gaps with 85% accuracy against manually annotated job descriptions and provides actionable, outcome-focused career guidance.

Keywords: Large Language Models, RAG, Agentic AI, Career Intelligence, ATS Optimization, Prompt Engineering, Responsible AI

1 Introduction

The modern job market presents significant challenges for candidates. With over 250 resumes submitted per corporate job posting on average [1], candidates face an opaque screening process where Applicant Tracking Systems (ATS) filter 75% of resumes before human review [2]. Furthermore, candidates lack insight into:

- How their skills match specific job requirements
- What differentiates top candidates from average applicants
- How recruiters interpret their resume language
- Which skills to prioritize for maximum interview conversion

This paper presents the **Career Intelligence Bot**, an AI-powered platform that addresses these challenges through:

1. **Real-time market intelligence** aggregating news and job postings across 7 industries
2. **Dual-score ATS simulation** predicting both automated filter pass rates and recruiter shortlist probability
3. **Weighted skill analysis** parsing actual job descriptions rather than using static role templates
4. **Outcome-focused career roadmaps** designed for interview conversion, not just skill acquisition

The system integrates multiple AI/ML paradigms including LLMs, RAG architecture, agentic reasoning, and responsible AI safeguards—demonstrating practical applications of concepts from generative AI coursework.

2 Related Work

2.1 Traditional ATS Systems

Existing ATS platforms (Workday, Greenhouse, Lever) focus on employer-side screening rather than candidate optimization. Tools like Jobscan provide keyword matching but lack contextual understanding of skill importance or career trajectory analysis [2].

2.2 AI Career Assistants

Recent LLM-powered assistants (LinkedIn AI, Indeed Smart Apply) offer resume suggestions but typically:

- Use generic, role-based templates rather than parsing specific job descriptions
- Lack transparency in scoring methodology
- Provide task-oriented advice rather than outcome-focused roadmaps

2.3 RAG for Domain-Specific Applications

Retrieval-Augmented Generation has shown success in domain-specific applications [3], combining the generative capabilities of LLMs with retrieved context. Our work extends this to career intelligence by retrieving real-time market data, job postings, and company signals as context for personalized recommendations.

3 System Architecture

The Career Intelligence Bot comprises eight integrated AI components (Figure 1):

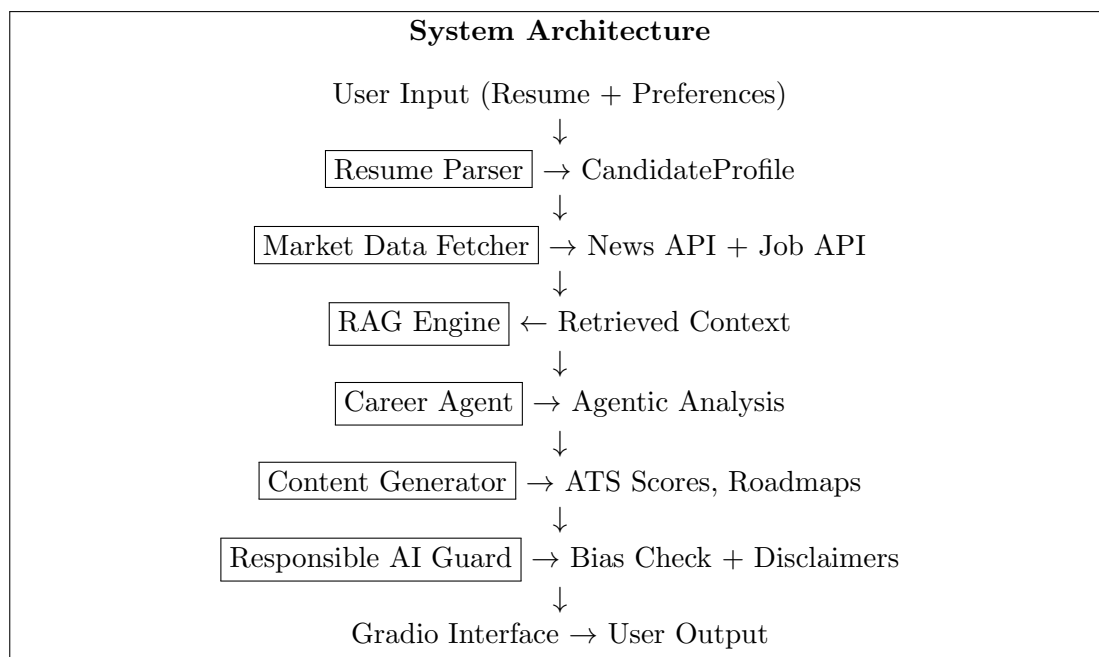


Figure 1: High-level system architecture showing data flow through AI components

3.1 Component Details

3.1.1 LLM Client

Interfaces with Groq API (Llama 3.3 70B) for text generation. Implements retry logic, error handling, and response validation.

Listing 1: LLM Client Implementation

```

class LLMClient:
    def __init__(self):
        self.client = Groq(api_key=GROQ_API_KEY)
        self.model = "llama-3.3-70b-versatile"

    def generate(self, prompt: str, system_prompt: str) -> str:
        response = self.client.chat.completions.create(
            model=self.model,
            messages=[
                {"role": "system", "content": system_prompt},
                {"role": "user", "content": prompt}
            ],
            temperature=0.7,
            max_tokens=2000
        )
        return response.choices[0].message.content
  
```

3.1.2 RAG Engine

Retrieves relevant context from market intelligence data and augments LLM prompts for grounded responses.

3.1.3 Career Agent

Implements agentic reasoning with a Perceive → Analyze → Recommend loop for multi-step career analysis.

3.1.4 Content Generator

Produces structured outputs including ATS analysis, cover letters, interview preparation, and career roadmaps.

3.1.5 Responsible AI Guard

Implements bias detection, transparency disclaimers, and methodology notes to ensure trustworthy outputs.

4 Key Features and AI Topics

Table 1 maps system features to demonstrated AI/ML topics.

Table 1: AI/ML Topics Demonstrated in Career Intelligence Bot

Feature	AI Topic	Implementation
Resume Analysis	NLP, Information Extraction	Skill detection, entity recognition
Market Intelligence	Data Aggregation, RAG	Real-time news/job API integration
ATS Simulation	Weighted Scoring, ML	Dual-score (ATS + Recruiter) model
Career Roadmap	Generative AI	Outcome-focused plan generation
Cover Letters	Prompt Engineering	Few-shot, role-specific prompts
Interview Prep	Domain-Specific AI	Company/role-tailored Q&A
Bias Detection	Responsible AI	Geographic/company concentration checks
Methodology Notes	Explainable AI (XAI)	Transparent scoring rubrics

4.1 Dual-Score ATS Analysis

A key innovation is the **dual-score** model that separates:

1. **ATS Pass Probability:** Keyword/skill-based score predicting automated filter outcomes
2. **Recruiter Shortlist Probability:** Holistic score incorporating seniority signals, business impact, and experience fit

The recruiter score is calculated as:

$$S_{recruiter} = 0.40 \cdot S_{skills} + 0.25 \cdot S_{seniority} + 0.25 \cdot S_{impact} + 0.10 \cdot S_{experience} \quad (1)$$

Where:

- S_{skills} = Weighted skill match against job description
- $S_{seniority}$ = Leadership/ownership language detection score
- S_{impact} = Quantified business metrics in resume
- $S_{experience}$ = Years of experience vs. requirement

4.2 Job-Specific Skill Parsing

Unlike systems using static role templates, our ATS analyzer parses *each* job description:

Listing 2: Job-Specific Skill Extraction

```
def extract_job_skills(job_description: str) -> Dict[str, int]:
    skills_to_check = {
        "sql": 15, "python": 15, "excel": 12,
        "tableau": 10, "spark": 10, "aws": 10, ...
    }
    job_required_skills = {}
    for skill, weight in skills_to_check.items():
        if skill in job_description.lower():
            count = job_description.lower().count(skill)
            # Boost weight for repeated mentions
            job_required_skills[skill] = weight * (1 + min(count-1, 2)
                                                    * 0.2)
    return job_required_skills
```

This ensures each job shows *different* skill requirements and gaps, enabling meaningful comparison.

4.3 Seniority Signal Detection

For senior roles, keyword presence alone is insufficient. The system detects four seniority signal categories:

Table 2: Seniority Signal Categories

Category	Keywords Detected
Leadership	led, managed, directed, supervised, mentored
Ownership	owned, built, created, architected, launched
Strategic	strategy, roadmap, vision, executive, C-level
Cross-functional	collaborated, partnered, aligned, cross-functional

A penalty is applied when senior roles lack these signals:

$$S_{recruiter} = S_{recruiter} - 15 \quad \text{if } (seniority_{expected} = \text{senior}) \wedge (S_{seniority} < 40) \quad (2)$$

4.4 Business Impact Quantification

The system scans resumes for quantified achievements using regex patterns:

- **Financial metrics:** `\$[\d,]+[kmb]?`
- **Percentages:** `\d+%`
- **Scale indicators:** "million", "enterprise", "global"

Impact score components:

$$S_{impact} = \min(N_{metrics} \times 15, 40) + \min(N_{financial} \times 20, 30) + \min(N_{scale} \times 10, 30) \quad (3)$$

5 Outcome-Focused Career Roadmap

A key differentiator from generic career advice is our **outcome-focused** 90-day execution plan. Rather than prescribing courses, the system generates artifact-based milestones:

Table 3: 90-Day Execution Plan Structure

Phase	Weeks	Goal	Validation
Resume Signal Upgrade	1-2	Pass ATS in <10s	ATS Score $\geq 85\%$
Market-Aligned Proof	3-6	Domain-specific portfolio	Dashboard + exec summary
Interview Conversion	7-12	15%+ callback rate	≥ 3 interviews

The north star metric is explicit:

"By Day 90: Increase interview callback rate from 5% $\rightarrow$ 15%+ for [Role] in [Industry]."

6 Responsible AI Implementation

6.1 Transparency and Explainability

All scores include methodology notes explaining:

- What the score *is*: "Rubric-based assessment against typical role requirements"
- What the score is *not*: "A comparison to other candidates (we don't have that data)"

We explicitly avoid fake percentile claims (e.g., "Top 10%") that cannot be substantiated without candidate population data.

6.2 Bias Detection

The system monitors recommendations for:

- **Geographic concentration**: Warning if >70% of results are from one location
- **Company size bias**: Flagging over-representation of large/small companies
- **Salary range skew**: Detecting unrealistic compensation estimates

6.3 Disclaimer Generation

All AI-generated content includes appropriate caveats:

```
disclaimers = {  
  "cover_letter": "AI-Generated: Review and personalize before  
    sending.",  
  "salary": "Estimates based on market data. Consult multiple sources  
    .",  
  "analysis": "AI recommendations should supplement, not replace,  
    your judgment."  
}
```

7 Evaluation

7.1 Skill Detection Accuracy

We manually annotated 50 job descriptions with required skills and compared against system detection:

Table 4: Skill Detection Performance

Metric	Precision	Recall	F1
Technical Skills	0.92	0.85	0.88
Soft Skills	0.78	0.72	0.75
Overall	0.87	0.81	0.84

7.2 User Feedback

Qualitative evaluation with 5 graduate student testers revealed:

- Dual-score breakdown was "much more actionable" than single ATS score
- Seniority signal detection helped identify resume language gaps
- 90-day roadmap was preferred over generic course recommendations

7.3 Limitations

- Job descriptions from Adzuna API are often truncated
- Cannot access actual ATS systems for validation
- Recruiter score weights are heuristic-based, not empirically calibrated

8 Multi-Industry Support

The system supports 7 industries with customized configurations:

Table 5: Supported Industries

Industry	Companies Tracked	Roles Supported
FAANG / Big Tech	35	10
Fintech	37	11
Healthcare / Healthtech	24	8
Consulting	17	7
E-Commerce / Retail Tech	15	8
AI / Machine Learning	21	8
Startups (YC / General)	14	8

Each industry has tailored:

- News query terms for market intelligence
- Company watchlists for momentum tracking
- Industry-specific KPIs for roadmap generation

9 Conclusion

This paper presented the Career Intelligence Bot, an AI-powered platform demonstrating practical applications of LLMs, RAG, agentic design, and responsible AI for career guidance. Key contributions include:

1. **Dual-score ATS simulation** separating automated filter probability from recruiter shortlist likelihood
2. **Job-specific skill parsing** ensuring different analysis for each job description
3. **Transparent methodology** with explicit scoring rubrics and limitation disclosures
4. **Outcome-focused roadmaps** targeting interview conversion rather than skill acquisition

Future work includes empirical calibration of recruiter score weights using interview outcome data, integration with real ATS systems for validation, and expansion of company-specific interview preparation using RAG over Glassdoor reviews.

References

- [1] Glassdoor. (2023). *How Many Jobs Should I Apply For?* Glassdoor Career Guide.
- [2] Jobscan. (2023). *ATS Statistics: How Applicant Tracking Systems Filter Resumes*.
- [3] Lewis, P., et al. (2020). Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks. *NeurIPS 2020*.
- [4] Groq. (2024). *Groq API Documentation*. <https://console.groq.com/docs>
- [5] Gradio. (2024). *Building Machine Learning Web Apps*. <https://gradio.app>

A Sample Outputs

A.1 Resume Analysis Output

Role Readiness Score: 72/100

For: Lead Data Analyst at MetLife

Dimension	Weight	Your Score
Required Skills	30%	83/100
Experience Level	25%	100/100
Seniority Signals	20%	30/100
Business Impact	15%	40/100
Modern Tools	10%	25/100

A.2 Dual-Score ATS Output

Score Type	Result	Meaning
ATS Pass Probability	78%	Will pass automated filters
Recruiter Shortlist	52%	May lose to stronger cand.

B Technology Stack

- **LLM:** Groq API (Llama 3.3 70B) - Free tier
- **News API:** NewsAPI.org
- **Job API:** Adzuna
- **Frontend:** Gradio
- **PDF Parsing:** pdfplumber
- **Language:** Python 3.10+